CAPSTONE PROJECT PROPOSAL DEFENSE

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BiasAperture: A Diagnostic Framework for Demographic Bias Auditing in Facial Analysis Systems

Bridging Fairness Research and Regulatory Governance via Reproducible, Standards-Aligned Auditing

Presenters

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Institutional Context & Supervision

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High Aggregate Accuracy Masks Severe Subgroup Failure

Buolamwini & Gebru (2018): Leading commercial gender-classification services exhibited high overall accuracy ($>$ 90%), but severe intersectional disparities:

- **Lighter Males:** **0.8% error rate**
- **Darker Females:** **34.7% error rate**
- **Disparity Gap:** Up to **43:1 error-rate ratio**

Facial analysis models are deployed in sensitive domains:

- **Biometric Access & Border Control:** Asymmetric false rejection rates causing wrongful profiling.
- **Identity Verification & KYC:** Demographic drop-off in automated banking onboarding.
- **Surveillance Systems:** Disproportionate false-positive matches for protected groups.

Distributional Masking

Standard empirical risk minimization (ERM) optimizes global sample loss:

$$\mathcal{R}(\theta) = \sum_{a} P(A = a)\, \mathcal{R}_a(\theta)$$

High aggregate metrics systematically conceal catastrophic failure modes in protected demographic slices.

The Industry Practice Failure

- Most audits remain **one-off, manual academic exercises**.
- Engineers lack a **reusable, vision-native software framework** for continuous bias regression testing.

From Academic Critique to Legal Compliance

EU AI Act (Regulation EU 2024/1689)

Phased Applicability via Article 113:

- **Article 10 (Data Governance):** Datasets must be examined for *"possible biases that are likely to lead to discrimination."*
- **Annex III Classification:** Biometric categorization inferring sensitive traits (race, age, gender) is classified as **High-Risk AI**.
- **Article 15 (Accuracy & Robustness):** Requires declared demographic performance metrics.

NIST AI RMF 1.0 (Measure Function)

Voluntary Measurement Taxonomy:

- **Measure 2.11:** Systematic measurement of bias and demographic parity.
- **Measure 1.1:** Documenting unquantifiable or low-sample risks ($n < 30$ guard).
- **Measure 1.3:** Software implementation cross-checking.

Compliance requires **reproducible, offline-auditable software infrastructure** translating legal mandates into concrete statistical metrics.

Comparative Landscape Audit: The Workflow Gap

Auditing Tool	CV Vision	Dual Backend	Support Guard	Proxy SHAP	Reg. Map	Offline Report
Fairlearn (Microsoft)	Tabular	Single	Partial	No	No	Partial
AIF360 (IBM)	Tabular	Single	No	No	No	No
Aequitas (Chicago)	Tabular	Single	Partial	No	No	Web/PDF
Google What-If Tool	Ad-hoc	Single	No	TensorBoard	No	Web-only
JFAM (Alvi et al.)	Vision	None	No	No	No	No
FAT Forensics	Tabular	Single	No	LIME/Kernel	No	Notebook
FairTest (Tramèr et al.)	Tabular	Single	Association tests	No	No	Text
BiasAperture (Ours)	**Vision**	**Dual (FL+AIF)**	$n \geq 30$ + **BCa**	**Targeted**	**Art. 10+RMF**	**Zero-Net HTML**

"There is an engineering gap between a mature body of fairness metrics and the availability of a reusable, standards-aligned software platform that operationalises this research into a repeatable auditing workflow for facial analysis systems."

Objectives & The Strict Diagnostic Boundary

General Objective

Design and build **BiasAperture**, a modular diagnostic software platform systematically identifying demographic accuracy disparities in facial analysis models and compiling regulator-legible reports.

5 Specific Objectives

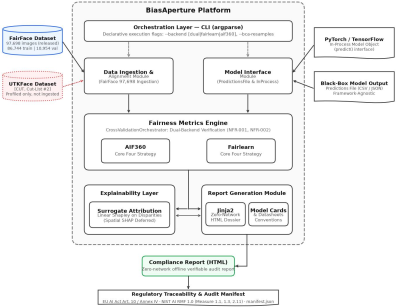
1. **Modular Ingestion:** Ingest datasets (FairFace) and prediction files (FR-001, FR-002).
2. **Dual-Backend Verification:** Cross-check Fairlearn & AIF360 with χ^2 and BCa bootstrap (FR-003, FR-004).
3. **Explainability Layer:** Provide targeted proxy attribution on flagged disparities ($p < 0.05$) (FR-005).
4. **Offline Report:** Generate zero-network HTML reports via Model Cards & Datasheets (FR-006).
5. **Regulatory Mapping:** Map metrics to EU AI Act Art. 10 and NIST AI RMF (FR-007).

Diagnostic integrity requires **separation of concerns**:

- **IN SCOPE:**
  - Data validation & profiling
  - Prediction file & model ingestion
  - Disparity metrics & statistical tests
  - Targeted proxy attribution
  - Compliance documentation
- **EXPLICITLY OUT OF SCOPE:**
  - Model retraining / fine-tuning
  - Loss function weight debiasing
  - Synthetic face image generation

Auditors must remain independent evaluators, never participants in model retraining.

Modular Pipeline Architecture



Five Cooperating Architectural Modules

1. **Data Ingestion & Alignment:** Reads FairFace; validates formats; maps labels into unified schema.
2. **Model Interface:** Framework-agnostic prediction ingestion (CSV/JSON or PyTorch model).
3. **Fairness Engine:** Strategy pattern over Fairlearn & AIF360; computes Core Four metrics + stats.
4. **Explainability Layer:** Selectively triggered proxy attribution on flagged disparities.
5. **Report Generator:** Flat Jinja2 compiler assembling zero-network offline HTML dossier.

- **Single Responsibility:** Each module owns one stage.
- **Open/Closed (Strategy):** Adding backends/metrics requires no engine modifications.
- **Dependency Inversion:** Pipeline depends on abstract interfaces, never concrete vendor libraries.

Design by Contract: The Locked Schema

Milestone M1 Contract Invariance

To enable concurrent multi-stream development without integration drag, data contracts were formally locked at Milestone M1:

- `SubjectRecord`: Primary datum holding normalized demographics and predictions:
  - `image_id: str`
  - `subgroup_race: RaceCategory` (7 categories)
  - `subgroup_gender: GenderCategory` (2 categories)
  - `subgroup_age: AgeCategory` (9 categories)
  - `true_label: int`, `predicted_label: int`
- `MetricResult`: Immutable metric payload with statistical confidence and governance tags.

Central limit theorems and asymptotic normality fail for small sample sizes ($n < 30$):

- Any demographic slice with $n < 30$ is strictly forbidden from publishing point estimates.
- **Enforced by** `MetricResult.__post_init__`:

$$\text{if } n < 30 \implies \begin{cases} \texttt{insufficient\_sample} = \text{True} \\ \texttt{metric\_value} = \text{None} \end{cases}$$

- Eliminates spurious, noisy conclusions in sparse intersectional cells (e.g., specific age $\times$ race intersections).

"Unreliable data must be explicitly flagged, never estimated unstably."

The Core Four Disparity Metrics

Metric	Mathematical Formulation	Fair Value	Parity Type
Demographic Parity Diff (DPD)	$\max_a P(\hat{Y} = 1 \mid A = a) - \min_a P(\hat{Y} = 1 \mid A = a)$	0.0	Selection
Disparate Impact Ratio (DIR)	$\min_a P(\hat{Y} = 1 \mid A = a) \;/\; \max_a P(\hat{Y} = 1 \mid A = a)$	1.0	Selection
Equal Opportunity Diff (EOP)	$\max_a \mathrm{TPR}_a - \min_a \mathrm{TPR}_a \quad (\text{conditioned on } Y = 1)$	0.0	Error Rate
Equalized Odds Diff (EOD)	$\max\left(\max_a \mathrm{TPR}_a - \min_a \mathrm{TPR}_a,\; \max_a \mathrm{FPR}_a - \min_a \mathrm{FPR}_a\right)$	0.0	Joint Error

Selection vs. Error-Rate Parity

- **DPD & DIR** measure independence: $\hat{Y} \perp A$.
- **EOP & EOD** measure separation: $\hat{Y} \perp A \mid Y$ (Hardt et al., 2016).

Multi-Class Policy: Macro-OvR Binarization

- Facial tasks involve multi-class targets (e.g. 9 age categories).
- We apply **Macro One-vs-Rest (OvR)** binarization for DPD, EOD, and EOP; DIR is evaluated per-class.

Heterogeneous Backend Harmonization

Reusing toolkits like Fairlearn and AIF360 out-of-the-box introduces hidden numerical divergence:

1. **EOD Divergence:**
   - Fairlearn computes **worst-case max gap**:
   $$\max(\Delta\mathrm{TPR}, \Delta\mathrm{FPR})$$
   - Native AIF360 computes **arithmetic mean**:
   $$\frac{1}{2}(\Delta\mathrm{TPR} + \Delta\mathrm{FPR})$$
2. **EOP Sign Mismatch:**
   - Fairlearn returns unsigned scalar ≥ 0.
   - AIF360 returns signed difference $\in [-1, 1]$ relative to privileged group.

Adapter abstraction guaranteeing cross-backend mathematical consistency:

- **EOD Harmonization:** Standardized to the **worst-case max gap** (Hardt et al., 2016) in both backends.
- **EOP Harmonization:** Normalized via absolute difference adapter ($|\mathrm{EOP}| \geq 0$).
- **Symmetric Bounded DIR:** Enforces symmetric bounded ratio:

$$\mathrm{DIR} = \min_a \mathrm{rate}_a \,/\, \max_a \mathrm{rate}_a \in [0, 1]$$

Why This Matters

Independent backends cross-validate each other: any divergence flags an implementation defect, not true disparity.

Three-Tier Statistical Inferential Defense

Sample-Size Invariants:

- Minimum global check: $n_a \geq 30$.
- Metric-specific cell counts:
  - **EOP:** $n_{Y=1,a} \geq 5$ (true positive support).
  - **EOD:** $n_{Y=1,a} \geq 5$ and $n_{Y=0,a} \geq 5$ (both positive and negative support).
- Fails trigger automatic: `insufficient_sample=True`

Tier 2: Hypothesis Testing

Contingency Testing & FWER:

- χ^2 test of independence against demographic distributions.
- **Fisher's Exact Test** fallback whenever expected cell frequency $E_{ij} < 5$.
- **Holm-Bonferroni Correction:** Controls Family-Wise Error Rate (FWER) across multi-group and intersectional hypothesis slices:

$$p_{(k)} \leq \frac{\alpha}{m - k + 1}$$

Confidence Intervals:

- **Bias-Corrected and Accelerated (BCa)** bootstrap ($B \geq 1,000$ resamples).
- Stratified within-subgroup resampling to hold observed n_a fixed.
- **Percentile Fallback Policy:**
  - Extreme acceleration $|a| > 0.5$.
  - Degenerate bias correction z_0.
  - Non-monotonic bounds.
- Replicate validity $\geq 90\%$.

"Disparity ratios and differences are effect sizes, not standalone proof of bias. They must always be contextualized by sample size, hypothesis testing, and rigorous confidence intervals."

Targeted Explainability & Proxy Attribution

Targeted Trigger Architecture

In high-throughput auditing, generating feature explanations for all images is computationally prohibitive.

- **Trigger Policy:** Attribution is computed **only** for demographic intersections flagged with **statistically significant disparities ($p < 0.05$ and $n \geq 30$)**.
- **Implementation:** Demographic surrogate attribution model quantifying feature sensitivity across demographic slices.
- **Skin Tone Colorimetry:** Individual Typology Angle (ITA):

$$\text{ITA} = \arctan\left(\frac{L^* - 50}{b^*}\right) \times \frac{180}{\pi}$$

Evaluates objective luminance and melanin pigmentation without racial proxy assumptions.

Impossibility Theorems for Feature Attribution:

- Feature attribution provides **exploratory proxy evidence**, **never causal proof** of algorithmic discrimination.
- Attribution maps reflect model sensitivity to correlated visual features (lighting, pose, facial geometry), which may or may not mediate disparate outcomes.

"BiasAperture reports explainability as supplementary diagnostic clues for model developers, explicitly demarcating correlation from causation."

Empirical Validation Benchmark: FairFace

FairFace Dataset (Kärkkäinen & Joo, 2021)

Balanced benchmark explicitly constructed to mitigate racial and demographic skew of prior datasets:

- **Dataset Scale on Disk:** **97,698 total labeled images** (86,744 training, 10,954 validation).
- **Demographic Taxonomy:**
  - **7 Races:** White, Black, Indian, East Asian, Southeast Asian, Middle Eastern, Latino.
  - **9 Age Groups:** 0–2, 3–9, 10–19, 20–29, 30–39, 40–49, 50–59, 60–69, 70+.
  - **2 Genders:** Male, Female.
- **Intersectional Slices:** $7 \times 9 \times 2 = 126$ potential demographic cells.

- **Architecture:** ResNet-34 multi-task classifier.
- **Pre-processing:** `dlib` 5-point alignment, 300×300 chips, 0.25 padding.
- **Validation Run:** Full inference executed over all **10,954 validation images** (`fairface_predictions_val.csv`).

- Initial proposal evaluated UTKFace as secondary benchmark.
- Falsified during data audit: UTKFace suffers severe **3-of-7 race collapse** and **noisy model-estimated age labels**.
- Descoping formally executed to protect audit integrity.

Empirical Benchmark Findings on 10,954 Images

Audited Model: FairFace ResNet-34 ($N = 10,954$)

Disparity Metric	Point Est.	95% BCa CI	Significance
Demographic Parity Diff	**0.923**	$[0.915, 0.930]$	$p = 0.000$ (**Fail**)
Disparate Impact Ratio	**0.070**	$[0.063, 0.078]$	$p = 0.000$ (**Fail**)
Equalized Odds Diff	*Flagged: Guarded on sparse cell* ($n_+ < 5$)		
Equal Opportunity Diff	*Flagged: Guarded on sparse cell* ($n_+ < 5$)		

- **Severe Selection Disparity:** DIR of 0.070 breaches the 0.80 four-fifths benchmark.
- **Narrow Confidence Bands:** High sample size ($N = 10,954$) yields tight BCa bounds, demonstrating that disparity is not sampling noise.

- When conditioning on $Y = 1$, sparse cells trigger the **Tier 1 Support Guard**.
- Rather than publishing wild numbers, BiasAperture marks them as **N/A (Guarded)**.
- Prevents misleading regulatory filings.

Every metric row links directly to:

- **EU AI Act:** Art. 10(2)(f) / 10(2)(g)
- **NIST AI RMF:** Measure 2.11 / 1.1

Standalone Offline Compliance Dossier

Dual Provenance Structure

Merges two leading AI documentation paradigms:

- **Model Cards (Mitchell et al., 2019):** Intended use cases, out-of-scope boundaries, demographic performance breakdown, ethical considerations.
- **Datasheets for Datasets (Gebru et al., 2018):** Dataset provenance, collection methodology, label noise analysis, demographic representation distributions.

Directly maps output sections to:

- **EU AI Act Art. 10(2):** Data governance & design choices.
- **Art. 10(3):** Statistical support & sample adequacy.
- **Art. 10(5):** Bias examination & testing.
- **NIST AI RMF:** Measure 2.11, 1.1, 1.3.

Auditing sensitive facial biometrics imposes strict privacy constraints:

- **Self-Contained Single File:** Compiled into a single `.html` file.
- **Zero External CDN Calls:** No external fonts, scripts, or tracker libraries.
- **Embedded Visuals:** All charts rendered via inline SVG or base64 data URIs.
- **Auditor Ready:** Can be securely emailed, archived, or inspected inside air-gapped corporate environments.

Verification Status

Generated reports (`audit_report_val_gender.html`) verified fully functional offline.

Single-Command Auditing Workflow

End-to-End CLI Invocation

```
# Run complete audit with proxy explainability:
uv run bias-aperture audit \
  --predictions data/processed/fairface_predictions_val.csv \
  --target-col service_test \
  --protected-axes race gender \
  --backend fairlearn --cross-validate \
  --explain \
  --output report/audit_report_val.html
```

1. **Schema Validation:** Ingests and verifies 10,954 rows.
2. **Dual Compute:** Runs Fairlearn + AIF360 strategies.
3. **Inference Engine:** Computes χ^2 and $B = 1,000$ BCa CIs.
4. **Explainability:** Evaluates surrogate attribution on disparities.
5. **Report Assembly:** Inlines SVG charts into offline HTML.

Terminal Output Telemetry

```
[INFO] Loaded 10,954 records from CSV.
[INFO] Slicing across ['race', 'gender']...
[INFO] Total demographic strata: 14.
[WARN] 0 strata flagged n < 30.
[INFO] Computing DPD, DIR, EOP, EOD...
[INFO] Cross-checking against AIF360Backend...
[SUCCESS] Dual-backend parity verified.
[INFO] Resampling B=1000 BCa bootstrap CIs...
[INFO] Significant disparity in 'Female' (p=0.000).
[INFO] Triggering surrogate attribution...
[SUCCESS] Report written -> audit_report_val.html
```

Direct drop-in integration into ML regression testing and CI/CD pipelines.

Verification, Contract Testing & Claim Ledger

55/55 Passing Pytest Suite

Strict test-driven verification across all core layers:

- **Unit Tests:** Schema typing, boundary invariant $n < 30$, CSV data loading.
- **Harmonization Tests:** Verifying Fairlearn vs. AIF360 parity on identical inputs.
- **Statistical Tests:** Validating χ^2, BCa resampling reproducibility, and fallbacks.
- **Reporting Tests:** Ensuring zero CDN dependencies and valid HTML formatting.

8-record ground-truth test suite verifies exact scalar calculations:

$$\text{DPD} = 0.500,\ \text{EOD} = 0.500,\ \text{DIR} = 0.3333$$

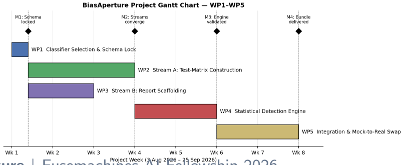
Documented ledger tracking foundational assumptions:

- **20 Active Research Claims:** Sourced from literature and verified in code.
- **5 Formally Falsified Hypotheses:**
  1. UTKFace secondary dataset viability $\longrightarrow$ **Falsified** (label noise & race collapse).
  2. AIF360 native EOD agreement $\longrightarrow$ **Falsified** (mean vs max divergence).
  3. Uniform intersectional sampling $\longrightarrow$ **Falsified** (sparsity in extreme age bins).
  4. Global SHAP compute feasibility $\longrightarrow$ **Falsified** (pivoted to targeted surrogate).
  5. Normal approximation for DIR CIs $\longrightarrow$ **Falsified** (boundary violation near 0).

Execution Schedule & 5-Tier Descoping Strategy

Milestone Progression & Timeline

- **M1 (Schema Lock):** Data models, taxonomy, contracts.
- **M2 (Scaffolding):** FairFace loader & HTML scaffolding.
- **M3 (Core Engine):** Metrics, harmonization, BCa bootstrap.
- **M4 (Integration):** CLI orchestrator & surrogate attribution.
- **M5 (Benchmark Audit):** FairFace ResNet-34 validation ($N = 10,954$).



Pre-defined contingency sequence protecting project delivery:

1. **Cut #1 (Web UI):** Retain robust, scriptable CLI.
2. **Cut #2 (UTKFace Dataset):** Retain FairFace as primary benchmark (**Formally executed**).
3. **Cut #3 (PDF Export):** Retain self-contained offline HTML dossier.
4. **Cut #4 (In-Process Models):** Retain CSV/JSON predictions file ingestion.
5. **Cut #5 (AIF360 Backend):** Retain Fairlearn backend (emergency fallback).

Ingestion + Fairness Engine + Statistical Rigor + Offline HTML Report are **never at risk**.

Summary of Core Engineering Contributions

1. Vision-Native Auditing Architecture

First open-source framework designed specifically for computer vision facial demographic auditing, handling multi-class and intersectional taxonomies without custom ad-hoc scripts.

2. Cross-Vendor Mathematical Harmonization

Identified and resolved fundamental metric divergences between Fairlearn and AIF360, establishing an auditable contract for cross-backend verification.

3. Three-Tier Inferential Defensibility

Codified empirical guards: $n \geq 30$ screening invariant, contingency testing with Holm-Bonferroni FWER control, and stratified vectorized BCa bootstrap confidence intervals.

4. Regulator-Legible Compliance Dossier

Built self-contained, zero-network HTML reports conforming to Model Cards and Datasheets, directly crosswalked to EU AI Act Article 10 and NIST AI RMF.

BiasAperture transforms theoretical fairness formulas into an automated, production-grade audit pipeline.

Thank You

BiasAperture: Demographic Bias Auditing in Facial Analysis Systems

We welcome questions and feedback from the examination panel.

Project Repositories & Artifacts

- **Source Code:** `github.com/fuseai-fellowship/...`
- **Verification Suite:** 55 passing tests (‘pytest‘)
- **Sample Audit Dossier:** `report/audit_report_val.html`

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Fusemachines AI Fellowship 2026

Backup Slide: The Novelty Defense

"Fairlearn and AIF360 already exist. Aren't you just writing a wrapper around existing open-source libraries?"

The Tooling Analogy

Docker did not invent cgroups or namespaces; Git did not invent diff algorithms. Their contribution was eliminating workflow friction, unifying disparate interfaces, and making capabilities reliably reusable.

Computer Vision Friction

Existing tools are tabular-first. Auditing computer vision models requires multi-class OvR binarization, face alignment, intersectional slicing, and image proxy attribution.

Toolkits natively disagree:

- Native Fairlearn & AIF360 calculate **divergent Equalized Odds** on identical data.
- Sign conventions for EOP conflict.
- Neither library provides $n < 30$ support guards, FWER-controlled χ^2 testing, or offline regulatory crosswalks.

Packaging these verified decisions into an auditable, reproducible workflow.

Backup Slide: Why Diagnostic-Only?

"Why didn't you build model debiasing or retraining algorithms into BiasAperture?"

1. Ethical Separation of Concerns

A diagnostic auditor must remain an independent evaluator. Folding mitigation or retraining into an auditing tool introduces an inherent **conflict of interest**:

- If an auditor automatically modifies weights to pass its own tests, the tests cease to be objective evaluations (Goodhart's Law).

2. Literature Precedent (Dehdashtian et al.)

Dehdashtian, Wang, & Boddeti (2024) formally partition fairness research into two distinct domains:

- **Diagnostic Auditing:** Measuring disparities, validating statistical confidence, documenting compliance.
- **Mitigation:** Pre-processing, in-processing adversarial training, post-processing threshold adjustments.

Downstream teams use BiasAperture reports to inform their own domain-specific mitigation strategies.

Backup Slide: Why BCa Bootstrap over Normal Approx.?

Failure of Standard Wald / Normal Approx.

Standard normal confidence intervals assume asymptotic symmetry:

$$\hat{\theta} \pm 1.96 \cdot \mathrm{SE}(\hat{\theta})$$

Why this fails for fairness auditing:

- **Ratio Metrics (DIR):** Ratio distributions min / max are heavily skewed with positive excess kurtosis.
- **Boundary Compression:** DPD, EOD, and EOP are bounded near $[0, 1]$. Normal approximation frequently leaks outside bounds (< 0 or > 1).

The Bias-Corrected and Accelerated bootstrap adjusts for:

1. **Bias Correction (z_0):** Adjusts for median bias between bootstrap replicates and empirical estimate:

$$z_0 = \Phi^{-1}\left(\frac{1}{B}\sum_{b=1}^{B} \mathbb{I}(\hat{\theta}_b^* < \hat{\theta})\right)$$

2. **Acceleration Parameter (a):** Corrects for rate of change of standard error via jackknife residuals.
3. **Transformation Invariance:** Respects theoretical bounds without artificial clipping.